

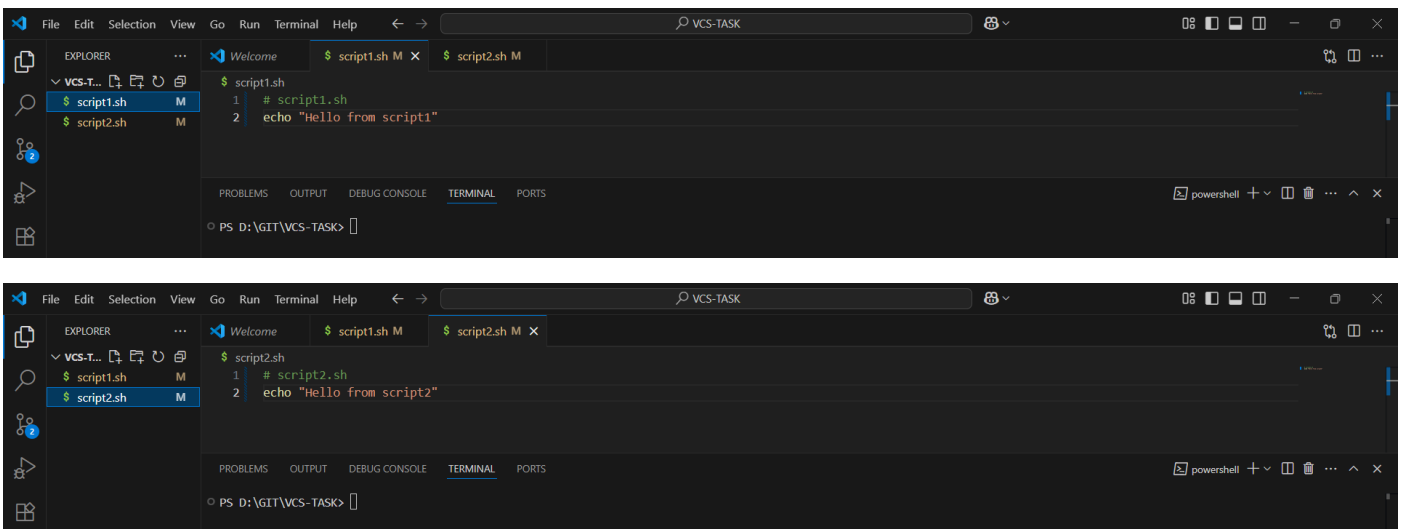
VCS Task

Task Description:

Establish a new directory, populate it with script files, initiate an empty repository on GitHub, convert the local directory into a Git repository, and link it to GitHub for pushing the code into the repository. Perform merge, rebase, stash commands in following github repo.

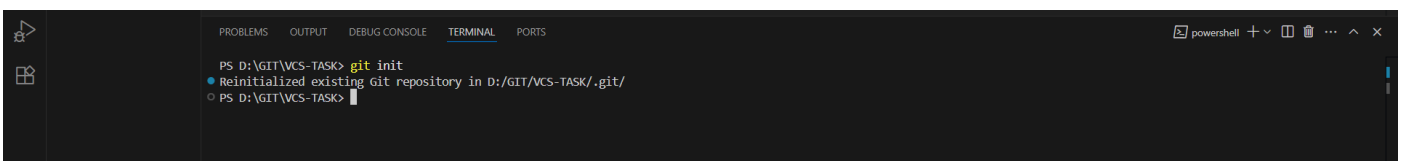
Solution:

✓ 1. Create Directory and Files



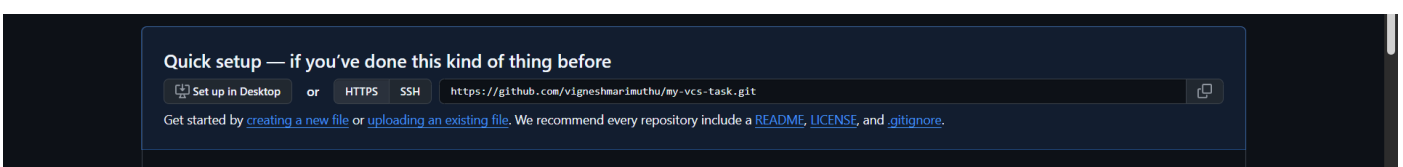
- Created a folder VCS-TASK and added two script files (script1.sh, script2.sh) using VS Code.

✓ 2. Initialize Git Repo



- Initialized Git using git init

✓ 3. Create GitHub Repo



- Created GitHub Repo <https://github.com/vigneshmarimuthu/my-vcs-task.git>

✓ 4. Connect Local Repo to GitHub

```
PS D:\GIT\VCS-TASK> git init
Reinitialized existing Git repository in D:/GIT/VCS-TASK/.git/
PS D:\GIT\VCS-TASK> git remote add origin https://github.com/vigneshmarimuthu/my-vcs-task.git
PS D:\GIT\VCS-TASK> git add .
PS D:\GIT\VCS-TASK> git commit -m "Initial commit with script1 and script2"
```

- Connected the Local repo to GitHub using `git remote add origin`.

✓ 5. Add, Commit & Push Files

```
PS D:\GIT\VCS-TASK> git remote add origin https://github.com/vigneshmarimuthu/my-vcs-task.git
PS D:\GIT\VCS-TASK> git add .
PS D:\GIT\VCS-TASK> git commit -m "Initial commit with script1 and script2"
[rebase-demo 1699711] Initial commit with script1 and script2
2 files changed, 0 insertions(+), 0 deletions(-)
PS D:\GIT\VCS-TASK> git branch -M main
PS D:\GIT\VCS-TASK> git push -u origin main
Enumerating objects: 16, done.
Counting objects: 100% (16/16), done.
Delta compression using up to 8 threads
Compressing objects: 100% (13/13), done.
Writing objects: 100% (16/16), 1.37 KiB | 701.00 KiB/s, done.
Total 16 (delta 2), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (2/2), done.
To https://github.com/vigneshmarimuthu/my-vcs-task.git
 * [new branch]      main -> main
branch 'main' set up to track 'origin/main'.
PS D:\GIT\VCS-TASK>
```

- Committed and pushed code to the main branch of my GitHub repository.

✓ 6. Git Branch & Merge

```
PS D:\GIT\VCS-TASK> git checkout -b feature-branch
Switched to a new branch 'feature-branch'
PS D:\GIT\VCS-TASK> echo "echo Feature branch update" >> script1.sh
PS D:\GIT\VCS-TASK> git commit -am "Updated script1 in feature-branch"
[feature-branch d749614] Updated script1 in feature-branch
1 file changed, 0 insertions(+), 0 deletions(-)
PS D:\GIT\VCS-TASK> git checkout main
Switched to branch 'main'
Your branch is up to date with 'origin/main'.
PS D:\GIT\VCS-TASK> git merge feature-branch
Updating 1699711..d749614
Fast-forward
 script1.sh | Bin 38 -> 94 bytes
1 file changed, 0 insertions(+), 0 deletions(-)
PS D:\GIT\VCS-TASK>
```

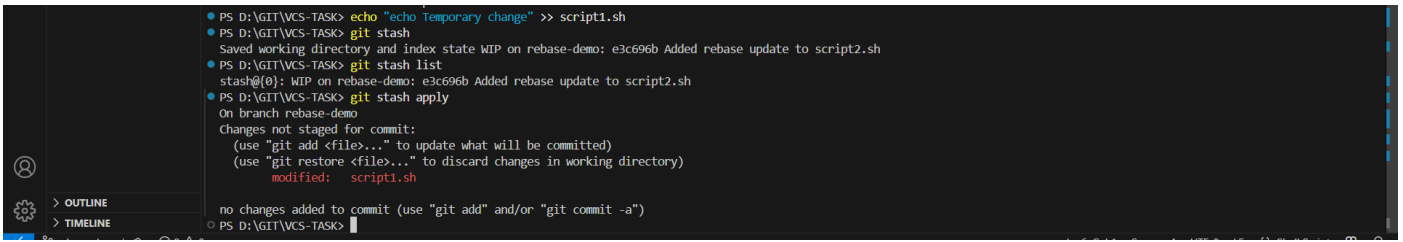
- Created a **feature-branch** to simulate changes and merged it into main using `git merge`.

✓ 7. Git Rebase

```
PS D:\GIT\VCS-TASK> git checkout -b rebase-demo
Switched to a new branch 'rebase-demo'
PS D:\GIT\VCS-TASK> echo "echo Rebase update" >> script2.sh
PS D:\GIT\VCS-TASK> git commit -am "Added rebase update to script2.sh"
[rebase-demo e3c696b] Added rebase update to script2.sh
1 file changed, 0 insertions(+), 0 deletions(-)
PS D:\GIT\VCS-TASK> git checkout main
Switched to branch 'main'
Your branch is ahead of 'origin/main' by 1 commit.
(use "git push" to publish your local commits)
PS D:\GIT\VCS-TASK> git checkout rebase-demo
Switched to branch 'rebase-demo'
PS D:\GIT\VCS-TASK> git rebase main
Current branch rebase-demo is up to date.
```

- Created a **rebase-demo** branch, added changes, and rebased it onto main using `git rebase`.

✓ 8. Git Stash



```
PS D:\GIT\VCS-TASK> echo "Temporary change" >> script1.sh
PS D:\GIT\VCS-TASK> git stash
Saved working directory and index state WIP on rebase-demo: e3c696b Added rebase update to script2.sh
PS D:\GIT\VCS-TASK> git stash list
stash@0: WIP on rebase-demo: e3c696b Added rebase update to script2.sh
PS D:\GIT\VCS-TASK> git stash apply
On branch rebase-demo
Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
        modified:   script1.sh
no changes added to commit (use "git add" and/or "git commit -a")
PS D:\GIT\VCS-TASK>
```

- Used git stash to save temporary changes and applied them back.

All steps were performed using **Visual Studio Code Terminal** and results were successfully verified on GitHub.
